

Non-determinism in LLM-as-Judge Graders: An Empirical Reproducibility Note

Author: Hiroki Tamba (Tamba Research Academy) · ORCID [0009-0004-7635-0741](https://orcid.org/0009-0004-7635-0741) **Status:** Preprint / technical note — **v1.1 (extended parameter coverage)** DOI: [10.5281/zenodo.20581781](https://doi.org/10.5281/zenodo.20581781) (concept · always latest) · v1.0 [10.5281/zenodo.20581782](https://doi.org/10.5281/zenodo.20581782) **Origin:** Generalized from a finding first reported as Japan-AISI/aisev issue #25 (2026-06-03).

Abstract

LLM-as-judge ("grader") components are now standard in evaluation harnesses, including safety evaluations where a pass/fail verdict gates downstream decisions. A widespread implicit assumption is that setting the grader's sampling temperature to 0 makes grading *deterministic* and therefore *reproducible*. This note shows that assumption is wrong on two levels.

Uncontrolled defaults silently inject randomness. In a real, publicly released evaluation harness (Japan AISI's aisev), the grader is invoked without setting temperature or seed. Because the OpenAI provider **omits** a `None` temperature from the request, the API applies its **default of 1.0** (not a deterministic default), so the grader effectively runs **unseeded at temperature 1.0**. Items near the grader's decision boundary then flip pass/fail across otherwise identical runs.

temperature=0 is necessary but not sufficient. Pinning the temperature to 0 removes the *rare* flips but not the *hard* ones: the actual grader (gpt-4o) still splits **2 of 7** items at temperature=0, and a Claude (Sonnet 4.6) grader independently reproduces instability. Determinism of the *grader call* is not guaranteed by temperature alone, and the effect is not provider-specific.

Other sampling parameters do not help. Restricting `top_p` (nucleus sampling) to 0.1 or forcing `top_k=1` (greedy decoding) leaves the same items unstable. The non-determinism originates before the sampling step. Meanwhile, at least one production model (Claude Opus 4.8) has deprecated temperature entirely, making the `temperature=0` mitigation inapplicable.

We provide a small reproduction harness, report measured per-item disagreement rates, and recommend concrete mitigations: set `temperature=0` explicitly where still supported, set a seed where the provider supports it, run multiple epochs and **report grader variance instead of a single point estimate**, and surface a grader-disagreement rate as a harness health metric.

1. Background

Modern evaluation pipelines increasingly replace human raters with an LLM "judge" that reads a model transcript and a rubric and emits a verdict (e.g. `correct` / `incorrect`, `pass` / `fail`). This is convenient and scalable, but it inherits a property practitioners often forget: **the judge is itself a stochastic model**. If the same transcript can be graded differently on two runs, then:

- benchmark scores are not reproducible,

- regression detection ("did this model get worse?") is confounded by grader noise, and
- safety gates that depend on a pass/fail threshold can flip for reasons unrelated to the model under test.

The common mitigation is "set temperature to 0." This note examines whether that mitigation is (a) actually applied in practice, and (b) sufficient when it is applied.

2. The discovered case: aisev

aisev is Japan AISI's open-source evaluation environment (Apache-2.0). All references below are to upstream commit `e0604d1e7997c2949e3cea40d219089dfd477fb4` (Japan-AISI/aisev main at 2026-04-16, the merge of PR #23: <https://github.com/Japan-AISI/aisev/commit/e0604d1e7997c2949e3cea40d219089dfd477fb4>). **This note does not redistribute aisev source code.** It describes the grader call path and points to the upstream files by path and commit hash, so each claim is verifiable against a fixed revision without vendoring any of the harness. Tracing the grader call path:

1. `scorer_provider.py` calls `model_graded_qa()` **without** passing `temperature` or `seed`.
2. This propagates into `inspect_ai`'s `GenerateConfig` with `temperature=None`.
3. The OpenAI provider **omits** a `None` temperature from the API request, so the API applies its **default of 1.0** — full sampling, not deterministic decoding.
4. Net effect: the grader runs **unseeded, at temperature 1.0**.

Nothing in the harness signals to the user that grading is being performed under maximal sampling noise. The pass/fail numbers *look* like fixed measurements.

3. Empirical measurement

Method. We hold a fixed set of 7 deliberately borderline question/answer pairs (see [src/repro_disagreement_rate.py](#)) and grade each item repeatedly under two configurations: the harness's **default** (temperature and seed left unset, so the OpenAI provider omits the parameter and the API applies its default of 1.0 — N = 20 runs per item) and an explicit **temperature=0** (N = 10 runs per item). The grading prompt and the grade-extraction regex mirror aisev's `get_graded_qa_scorer` and `inspect_ai`'s default `model_graded_qa` instructions verbatim, so the verdict (C/I) is parsed exactly as aisev parses it. We compute the per-item *disagreement rate* — the fraction of runs whose verdict differs from the item's majority verdict.

Results — OpenAI gpt-4o grader (aisev's default). Three regimes appear: *stable* (no disagreement), *rare* (a single flip), and *strong* (near 50/50).

Item	default temp, N=20	temperature=0, N=10	Regime (default)	Disagreement (default)
item1	20 I / 0 C	10 I	stable	0.00
item2	20 I / 0 C	10 I	stable	0.00

item3	20 C / 0 I	10 C	stable	0.00
item4	11 I / 9 C	6 I / 4 C	strong	0.45
item5	19 I / 1 C	10 I	rare	0.05
item6	12 C / 8 I	9 C / 1 I	strong	0.40
item7	19 C / 1 I	10 C	rare	0.05

(Grader gpt-4o, aisev's default; the exact dated model snapshot was not captured in the run log. Raw outputs: [data/repro_raw_openai.json](#).)

Under the default configuration, **4 of 7 items (4, 5, 6, 7) are non-reproducible** — two by a coin-flip margin (item4, item6) and two by a single flip (item5, item7). For these items the reported pass/fail is not a property of the answer under test; it is a draw from the grader's sampling distribution.

Pinning **temperature=0 helps but does not close the gap**. It stabilises the two *rare* items (item5 and item7 become unanimous), yet the two *strong* items still flip — item4 splits 6/4 and item6 still yields a dissenting verdict (9/1) over just 10 runs. So on the actual aisev grader, temperature=0 reduces non-reproducibility from **4/7 to 2/7**: necessary, but not sufficient.

4. The effect is not provider-specific

A natural objection is that §3 captures a quirk of one provider's grader. It does not. Running the identical 7-item protocol with a **Claude (Sonnet 4.6)** grader reproduces the same qualitative picture: 2 of 7 items are non-reproducible at the default temperature, and **still 2 of 7 at temperature=0** (item6: 8 I / 2 C; item7: 9 C / 1 I, over 10 runs).¹ The *specific* unstable items differ between graders — OpenAI's temperature=0 instability falls on items 4 and 6, Claude's on items 6 and 7 — but the conclusion is identical: pinning the temperature does not make grading deterministic.

This falsifies the operational belief that temperature=0 $\Rightarrow$ deterministic grading. Residual non-determinism in production LLM serving can arise from sources unrelated to the sampling temperature, including:

- batching / request-coalescing effects on the serving side,
- mixture-of-experts routing variability,
- non-associative floating-point reductions across hardware/kernels,
- provider-side load balancing across non-identical model replicas.

The practical consequence: **temperature=0 reduces, but does not eliminate, grader disagreement**. Reproducibility must be *measured*, not assumed.

4a. Sampling parameter coverage (v1.1)

§4 showed that residual non-determinism is not provider-specific but tested only temperature. v1.1 asks: can other sampling parameters — top_p (nucleus sampling) or top_k — eliminate the remaining flips? It also extends the cross-model check to additional Claude tiers.

Method. Same 7-item protocol as §3, temperature=0 throughout. Seven conditions: three OpenAI (gpt-4o with top_p ∈ {default, 0.1, 0.5}), four Anthropic (Sonnet 4.6, Opus 4.8, Haiku 4.5 at defaults; Sonnet 4.6 with top_k=1). N = 10 runs per item per condition (490 API calls total). Raw outputs: [data/repro_raw_v11_extended.json](#) (OpenAI, 210 calls), [data/repro_raw_v11_anthropic.json](#) (Anthropic, 280 calls).

Results.

Condition	Model	top_p	top_k	Non-repro	Unstable items
baseline (§3 replication)	gpt-4o	—	—	2/7	4, 6
+ top_p=0.1	gpt-4o	0.1	—	2/7	4, 6
+ top_p=0.5	gpt-4o	0.5	—	2/7	4, 6
baseline	claude-sonnet-4-6	—	—	1/7	6
—	claude-opus-4-8	—	—	ERROR	(see below)
—	claude-haiku-4-5-20251001	—	—	1/7	6
+ top_k=1	claude-sonnet-4-6	—	1	1/7	6

Four findings emerge:

1. top_p is not a mitigation. Restricting the nucleus from the default to 0.1 (the tightest practical setting) does not change which items flip or how often. All three OpenAI conditions produce the same 2/7 pattern on the same items (4 and 6). The instability is in the top token's logit, not in the tail of the distribution that top_p trims.

2. top_k=1 (forced greedy) does not eliminate flips. On Sonnet 4.6, forcing the decoder to select only the single highest-probability token still produces 1/7 non-reproducibility (item 6: 8 I / 2 C — slightly worse than the baseline 9 I / 1 C). This is strong evidence that the non-determinism originates *before* the sampling step, in the forward pass itself (floating-point non-associativity, batch-dependent numerics, or MoE routing).

3. Non-determinism is model-tier-independent. Both Sonnet 4.6 and Haiku 4.5 flip the same item (item 6) at the same rate (9/1). The effect is not an artifact of a particular model size or architecture tier within a provider.

4. Sampling control is being removed. Claude Opus 4.7 and 4.8 reject explicit temperature values in the range [0, 1) with HTTP 400 ("temperature is deprecated for this model"); only temperature=1.0 (the default, full sampling) is accepted. The gate is on the *value*, not on the parameter being present. Furthermore, top_p and top_k are also deprecated on these two models — the entire sampling-control class is gone, not just temperature=0.² This means the primary recommendation of §6 — "set temperature=0 explicitly" — is *already inapplicable* to these models, and there is no alternative sampling-side knob to reach for. As providers move toward models with fixed or internally managed decoding, sampling-parameter mitigations will become progressively less

available, strengthening the case for the structural mitigations in §6 items 3–5 (epochs, variance reporting, and health metrics).

Note on temporal stability. The v1.0 Claude Sonnet 4.6 baseline (§4) found 2/7 non-reproducible items (items 6 and 7); the v1.1 re-run finds 1/7 (item 6 only, item 7 now unanimous at 10 C / 0 C). At $N = 10$ this is consistent with sampling noise (v1.0 item 7 was 9 C / 1 I — a 10% flip rate can easily produce 0 flips in 10 trials). It may also reflect serving-infrastructure changes between measurement dates. Either way, it underscores that a single measurement pass does not characterize the grader's long-term stability.

5. Why this matters

For ordinary capability benchmarks, grader noise inflates variance and can change rankings of closely-spaced models. For **safety** evaluations the stakes are higher: a pass/fail verdict near the boundary may gate a deployment decision, a red-team sign-off, or a public claim about a guardrail's coverage. If that verdict is a coin flip, the evaluation is reporting noise as a safety property.

6. Recommended mitigations

1. **Set `temperature=0` explicitly** at the grader call site, where the provider still supports it. Never rely on the provider's `None` default — it may mean `1.0`. Note that some models (e.g. Claude Opus 4.8) have deprecated `temperature` entirely (§4a).
2. **Set `seed`** where the provider supports it, and record it in the run metadata.
3. **Run `epochs > 1`** for the grader and **report variance / a confidence interval**, not a single point estimate.
4. **Treat near-boundary items as a first-class signal.** Surface a per-run *grader-disagreement rate* as a harness health metric; a high rate means the score is not trustworthy regardless of its central value.
5. **Log the effective grader config** (model, temperature, seed) into the results artifact so that "what temperature was the judge at?" is answerable after the fact.

7. Reproduction

The reproduction harness is [src/repro_disagreement_rate.py](#) (v1.0 protocol) and [src/repro_v11_extended.py](#) (v1.1 extended parameter sweep). Both mirror `aisev`'s grading prompt and `inspect_ai`'s grade-extraction regex verbatim. The raw grader outputs backing §3–§4a are in [data/](#):

- `repro_raw_openai.json` — v1.0 OpenAI results (§3)
- `repro_raw_anthropic.json` — v1.0 Anthropic results (§4)
- `repro_raw_v11_extended.json` — v1.1 OpenAI `top_p` sweep (§4a, 210 calls)
- `repro_raw_v11_anthropic.json` — v1.1 Anthropic multi-model + `top_k` (§4a, 280 calls)

See [README.md](#) for setup and the exact run counts.

8. Related work and positioning

LLM-as-judge reliability. Using an LLM to score model outputs is now standard practice (Zheng et al., 2023). Most of this literature characterizes *systematic* biases of the judge — position bias, verbosity bias, and self-preference (Wang et al., 2023) — that is, skews that persist *across* runs. The finding here is on an orthogonal axis: *run-to-run* non-determinism, where the same judge returns different verdicts for the identical (transcript, rubric) pair on repeated calls. The two concerns are complementary: a debiased judge can still be irreproducible, and a perfectly reproducible judge can still be biased.

Eval-harness grader configuration. Frameworks such as Inspect AI (UK AISI) expose the grader as a configurable model call (`model_graded_qa`, `GenerateConfig`), including `temperature` and, where the provider supports it, `seed`. The gap this note highlights is not in a framework's capabilities but in its *defaults*: when these knobs are left unset, the underlying provider may silently sample (OpenAI maps `temperature=None` → `1.0`), so a harness can report grades produced under full sampling noise without ever signalling it. This is a configuration-level reproducibility hazard, not a modeling one.

Residual non-determinism at `temperature=0`. Even under greedy decoding, production LLM serving is generally not bit-reproducible: request batching, mixture-of-experts routing, and non-associative floating-point reductions across kernels and hardware can change the argmax for borderline tokens; recent analysis attributes `temperature=0` nondeterminism primarily to a lack of *batch invariance* in core kernels, where numerics shift with batch size and sequence slicing (He et al., 2025). The cross-model check in §4 is consistent with this — a Claude grader at `temperature=0` still flipped 2/7 items — and motivates the operative recommendation of this note: *measure grader disagreement; do not assume it away*.

Pointers. - Original report: Japan-AISI/aisev issue #25 — "*Reproducibility Improvement Proposal — Grader's Pass/Fail Fluctuates Due to Uncontrolled Temperature/Seed.*" <https://github.com/Japan-AISI/aisev/issues/25> - Inspect AI (UK AISI): `GenerateConfig`, `model_graded_qa`.

References.

1. L. Zheng, W.-L. Chiang, Y. Sheng, et al. "Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena." *Advances in Neural Information Processing Systems 36 (NeurIPS 2023), Datasets and Benchmarks Track*, 2023. arXiv:2306.05685, DOI: 10.48550/arXiv.2306.05685.
2. P. Wang, L. Li, L. Chen, et al. "Large Language Models are not Fair Evaluators." 2023. arXiv:2305.17926, DOI: 10.48550/arXiv.2305.17926.
3. H. He et al. "Defeating Nondeterminism in LLM Inference." Thinking Machines Lab, 2025. <https://thinkingmachines.ai/blog/defeating-nondeterminism-in-llm-inference/>

Scope of this deposit

To stay within upstream licensing and good-citizen norms, this deposit contains **only the author's own artifacts**:

- the reproduction harness (`src/`),

- the measured results and their analysis (this note), and
- a written description of the upstream grader call path that references `Japan-AISI/aisev` by **file path and commit hash** rather than copying its source.

It does **not** redistribute any portion of the `aisev` codebase. Readers reproduce the upstream context by checking out the referenced commit of the upstream repository directly.

Versioning

This artifact follows a two-version plan, archived under a single Zenodo *concept* DOI with a distinct *version* DOI per release:

- **v1.0 — report.** Documents the finding and the measured non-determinism as of upstream commit `e0604d1` (main at 2026-04-16, PR #23), before any fix is adopted.
- **v1.1 (this release) — extended parameter coverage.** Adds `top_p` sweep (OpenAI), multi-model check (Claude Sonnet / Opus / Haiku), `top_k=1` forced-greedy check, and the Opus 4.8 temperature-deprecated finding. Contributor credit: Eirik Botten Nicolaysen.
- **v2.0 (planned) — resolution.** If the proposed mitigations are adopted upstream, v2 will add a short "Resolution" section recording the change (the upstream PR / commit and the post-fix disagreement rate). Citations to the concept DOI will resolve to the latest version; citations to a version DOI remain pinned.

See [CHANGELOG.md](#).

Acknowledgments and tooling

The empirical finding, reproduction code, and analysis are the author's own. Eirik Botten Nicolaysen (avalysset / EcoDeco AS; ORCID [0009-0001-9188-6788](#)) independently verified the non-determinism on a separate infrastructure and contributed observations on `top_p/top_k` coverage and cross-model scoping that informed v1.1. Drafting of this note was assisted by an LLM (Claude); all claims and data were verified by the author.

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Machine-readable metadata: [CITATION.cff](#).

Cross-model check: grader `claude-sonnet-4-6`, same 7-item set, $N = 20$ at the default temperature and $N = 10$ at `temperature=0`. Non-reproducible items were {6, 7} under both conditions (default — item6 14 I / 6 C, item7 15 C / 5 I; `temperature=0` — item6 8 I / 2 C, item7 9 C / 1 I). The `temperature=0` batch is small and

is meant only to show that disagreement does not vanish, not to estimate a precise rate. Raw outputs: [data/repro_raw_anthropic.json](#). ■

The deprecation is specific to Opus 4.7 and 4.8. Sonnet 4.6, Haiku 4.5, and earlier Opus versions (4.5, 4.6) accept explicit `temperature`, `top_p`, and `top_k` normally. Independent verification by Eirik Botten Nicolaysen (avalysset): <https://gist.github.com/avalysset/59a81e2961d45d4ba76de56d592b1110>. ■